ECON 8185-002 Homework 5

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December 13, 2022

Aiyagari's (1994) Incomplete Markets Model

Consumer's Problem

Let us begin by specifying the recursive problem of a consumer in the standard incomplete markets model in a partial equilibrium:

$$\begin{aligned} V(a,s) = &\max_{c,a'} \ \{u(c) + \beta \mathbb{E}[V(a',s')|s]\} \\ \text{s.t.} \quad & c + a' \leq Ra + wl(s) \\ & c \geq 0, \ a' \geq -\phi \end{aligned}$$

It helps to redefine the problem in terms of cash in hands $x = Ra + wl(s)$ as the state variable. We have:

$$\begin{aligned} V(x,s) = &\max_{c,a'} \ \{u(c) + \beta \mathbb{E}[V(x'(s'),s')|s]\} \\ \text{s.t.} \quad & c + a' \leq x \\ & c \geq 0, \ a' \geq -\phi \\ & x'(s') = Ra' + wl(s') \quad \forall s' \end{aligned}$$

Next, let us redefine the problem by normalizing the borrowing constraint to zero.

$$\begin{aligned} \hat{a}' &= a' + \phi \\ z &= x + \phi \\ z'(s') &= R\hat{a}' + wl(s') - (R-1)\phi \end{aligned}$$

Then, finally, substituting out consumption, the recursive problem of the consumer can be expressed as:

$$\begin{aligned} V(z,s) = &\max_{0 \leq \hat{a}' \leq z} \ \{u(z - \hat{a}') + \beta \mathbb{E}[V(z'(s'),s')|s]\} \\ \text{s.t.} \quad & z'(s') = R\hat{a}' + wl(s') - (R-1)\phi \quad \forall s' \end{aligned}$$

Production Technology

A representative firm with constant returns to scale technology, $F(K,L) = AK^{\alpha}L^{1-\alpha}$ chooses K, L to maximize profits:

$$\begin{aligned} w &= (1-\alpha)AK^{\alpha}L^{-\alpha} \\ R - 1 + \delta &= \alpha AK^{\alpha-1}L^{1-\alpha} \end{aligned}$$

To define a stationary general equilibrium, we impose market clearing conditions.

Stationary General Equilibrium
A stationary equilibrium is defined by:

- Prices $R,\ w(R)$
- A value function $V : A \times S \to \mathbb{R}$
- Policy functions $g : A \times S \to A,\ g_c : A \times S \to \mathbb{R}_+$

such that:

- Given prices, the policy functions solve the consumer's problem, and the value function attains the maximum.
- Given prices, the firm chooses K, L to maximize profits:

$$w(R) = (1-\alpha)AK^{\alpha}L^{-\alpha}$$
$$R - 1 + \delta = \alpha AK^{\alpha-1}L^{1-\alpha}$$

- A Stationary Distribution Γ of wealth and employment states is implied by the policy functions.
- Markets Clear:

$$K = \iint g(a,s)d\Gamma$$
$$L = \iint l(s)d\Gamma$$
$$\iint g_c(a,s)d\Gamma + K = AK^{\alpha}L^{1-\alpha} + (1-\delta)K$$

Solving for the Stationary equilibrium
Using Firm's FOCs, define:

$$K(R) = L\left(\frac{\alpha A}{R-1+\delta}\right)^{\frac{1}{1-\alpha}}$$
$$w(R) = (1-\alpha)A^{\frac{1}{1-\alpha}}\left(\frac{\alpha}{R-1+\delta}\right)^{\frac{\alpha}{1-\alpha}}$$

For the consumer problem, use the power utility function:

$$u(c) = \frac{c^{1-\gamma}}{1-\gamma}$$

The Euler equation from the consumer's problem is given by:

$$(z - \hat{a}')^{-\gamma} \geq \beta R\ \mathbb{E}[(z' - \hat{a}'')^{-\gamma}]$$

As in Aiyagari (1994), I assume the labour endowment process is given by:

$$\ln l_{t+1} = \varphi \ln l_t + \varepsilon_{t+1}, \quad \epsilon_t \sim \mathcal{N}(0, \sigma^2(1-\varphi^2))$$

where φ is the AR(1) coefficient and σ is the coefficient of variation of hours worked. I discretize the endowment process to a 19 state Markov chain using Rouwenhorst's method as in Kockecky and Suen (2010).

I solve for the assets policy function $g(a,s)$ by iterating on the Euler equation using the endogenous grid method. The EGM algorithm can be summarized as:

1. Set a grid for $\hat{a}'$. Compute $z'(s') = R\hat{a}' + y(s')$.
2. Guess an initial policy function for assets tomorrow $\hat{a}''$.
3. Find a z consistent with $\hat{a}'$ that solves the Euler equation, given the guess for policy function.
4. Update the policy function by interpolating on the exogenous grid z'.
5. Check tolerance criteria. If tolerance criteria is met, we have found the assets policy function. Otherwise, go back to step 3.

Finally, to obtain the stationary distribution, I simulate the economy for $N = 50,000$ and $T = 500$ and compute the average savings, $\bar{A}(R)$ at the end of the time period. An outer loop iterates over a grid of R and computes $K(R) - \bar{A}(R)$. Finally, using interpolation, I solve for R^* that sets $K(R) - \bar{A}(R) = 0$. The model parameterization is as follows:

Parameter	Value
β	0.96
γ	2
ϕ	0
α	0.36
δ	0.08
A	1
φ	0.95
σ	0.2

Results: CRRA Preferences

For the baseline model with $\gamma = 2$, Figure 1 shows the investment/savings demand and supply curves that determine R^*, analogous to Figure IIb in Aiyagari (1994). Figure 2 shows the Stationary marginal wealth density and distribution. Figures 3 and 4 show the same plots for $\gamma = 10$. Table 1 shows the equilibrium values and statistics for both $\gamma = 2, 10$. Increasing γ induces significantly higher savings (by shifting the savings curve downwards) and the equilibrium interest rate falls. However, since coefficient of relative risk aversion as well as the elasticity of intertemporal substitution (EIS) are changing by changing γ, it it not clear which effect induces more savings (and if both do, we cannot decompose the individual effects since both RRA and EIS are varying together).

Table 1: CRRA Preferences: Stationary Equilibrium Results

Statistic	$\gamma = 2$	$\gamma = 10$
R^*	1.038038	1.013260
$W(R)$	1.198360	1.368193
$K(R)$	5.710673	8.252325
$\bar{A}(R)$	5.710673	8.252325
L	1	1
Y	1.872438	2.137802
C	1.415584	1.477616
Borrowing constrained population (%)	6.05	0.36
Wealth share of top 1% (%)	5.31	3.93
Gini Coefficient	0.55	0.41

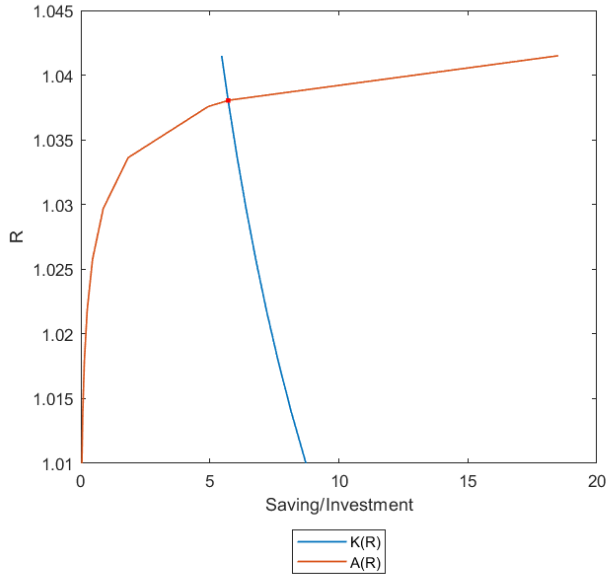

Figure 1: Equilibrium Determination, $\gamma = 2$

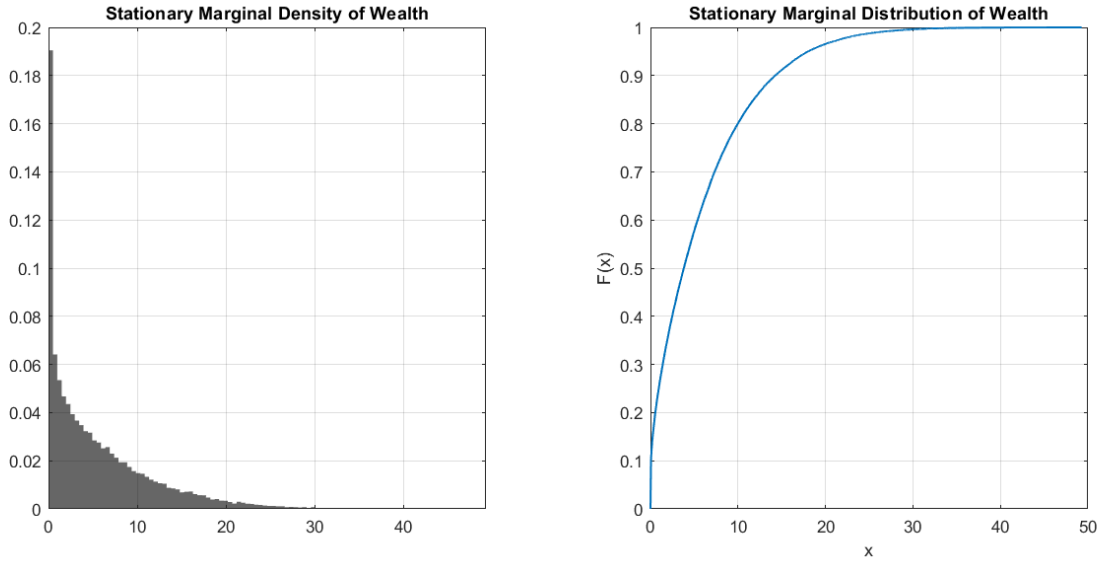

Figure 2: Stationary Marginal Distribution of Wealth, $\gamma = 2$

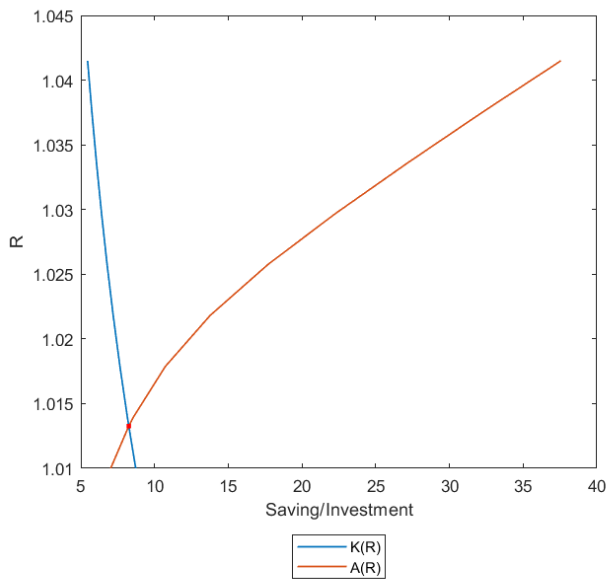

Figure 3: Equilibrium Determination, $\gamma = 10$

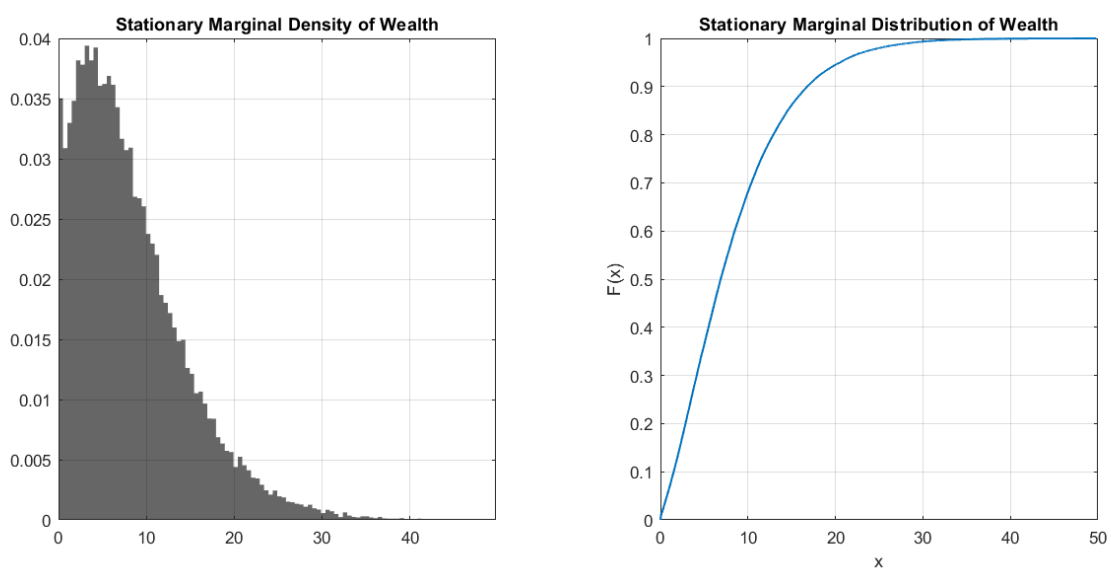

Figure 4: Stationary Marginal Distribution of Wealth, $\gamma = 10$

Results: Epsten-Zin Preferences

I use the following specification of Epsten-Zin (EPZ) preferences for solving the model:

$$V_t = \left[(1-\beta)c_t^{\rho} + \beta[\mathbb{E}V_{t+1}^{1-\gamma}]^{\frac{\rho}{1-\gamma}}\right]^{\frac{1}{\rho}}$$

The first order condition is given by:

$$(1-\beta)(z-\hat{a}')^{\rho-1} \geq \beta R[\mathbb{E}V(z',s')^{1-\gamma}]^{\frac{\rho}{1-\gamma}-1}\mathbb{E}[V(z',s')^{-\gamma}V_z(z',s')]$$

I solve for the policy function of assets from the above first order condition by iterating on the value function using the endogenous grid method.

First, let us fix EIS $= 0.5$ ($\rho = -1$) and vary γ. Table 2 shows the results and figures 5 and 6 show the stationary distributions for $\gamma = 2, 10$ respectively. Increasing RRA while holding EIS constant decreases the equilibrium interest rate and equilibrium savings increase, but compared to the CRRA case, the change is much smaller now. This suggests that change in EIS has a stronger effect on the interest rate and savings and since we did not allow EIS to change here, the effect is much smaller.

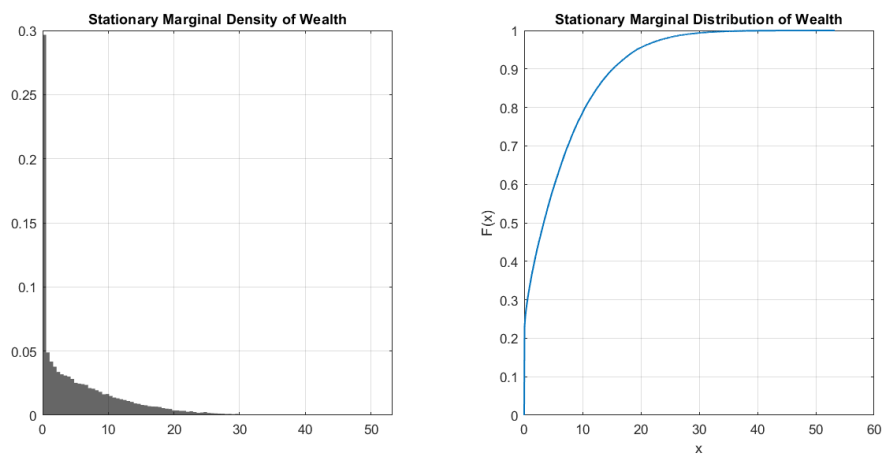

Figure 5: Stationary Marginal Distribution of Wealth, $\gamma = 2$, EIS $= 0.5$

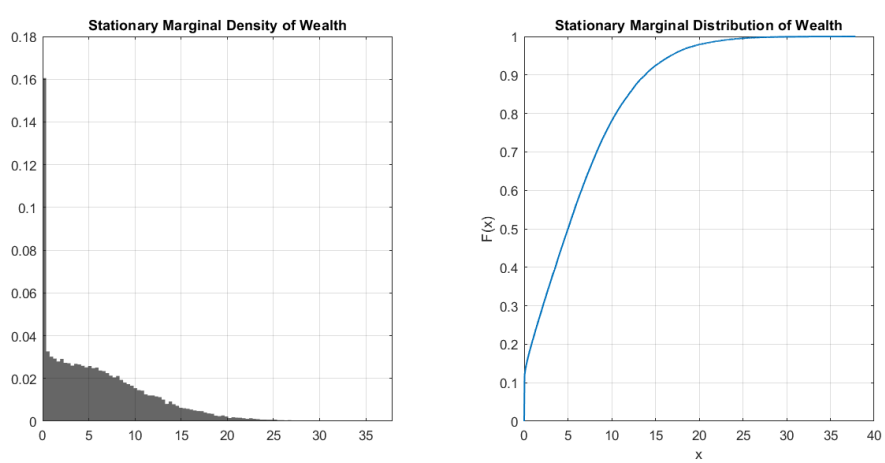

Figure 6: Stationary Marginal Distribution of Wealth, $\gamma = 10$, EIS $= 0.5$

Table 2: EPZ preferences: Stationary Equilibrium Results, EIS =0.5

Statistic	$\gamma = 2$	$\gamma = 10$
R^*	1.038468	1.032982
$W(R)$	1.195913	1.228235
$K(R)$	5.678340	6.114952
$\bar{A}(R)$	5.678340	6.114952
L	1	1
Y	1.868614	1.919117
C	1.414347	1.429921
Borrowing constrained population (%)	16.71	8.09
Wealth share of top 1% (%)	5.72	4.21
Gini Coefficient	0.60	0.49

Next, let us fix RRA = 2 and vary EIS in the range $\{0.1, 0.2, 0.5, 0.8, 2\}$. Table 3 shows the results and figures 7-11 give the distribution plots. Note that the baseline case is EIS=0.5. Decreasing EIS to 0.1 induces a strong decline in the interest rate and an increase in equilibrium savings. This change is much larger than what we observed by changing RRA, holding EIS constant. However, both these changes are smaller than what we observe in the case of CRRA preferences. This implies that the effect of changing γ from 2 to 10 in CRRA preferences can be decomposed into an EIS effect and a RRA effect. While these effects are implicit in CRRA preferences, EPZ preferences allow us to decompose them. We observe that the EIS effect is much stronger. Furthermore, Table 3 shows that an increase in EIS reduces savings and increases the equilibrium interest rate.

Table 3: EPZ preferences: Stationary Equilibrium Results, RRA=2

Statistic	**EIS=0.1**	**EIS=0.2**	**EIS=0.5**	**EIS=0.8**	**EIS=2**
R^*	1.029534	1.034790	1.038468	1.039450	1.040455
$W(R)$	1.249840	1.217318	1.195913	1.190372	1.184778
$K(R)$	6.418427	5.965159	5.678340	5.605560	5.532681
$\bar{A}(R)$	6.418427	5.965159	5.678340	5.605560	5.532681
L	1	1	1	1	1
Y	1.952874	1.902059	1.868614	1.859957	1.851215
C	1.439400	1.424846	1.414347	1.411512	1.408600
Borrowing constrained population (%)	18.78	18.47	16.71	15.85	14.14
Wealth share of top 1% (%)	6.14	5.95	5.72	5.56	5.09
Gini Coefficient	0.62	0.61	0.60	0.59	0.56

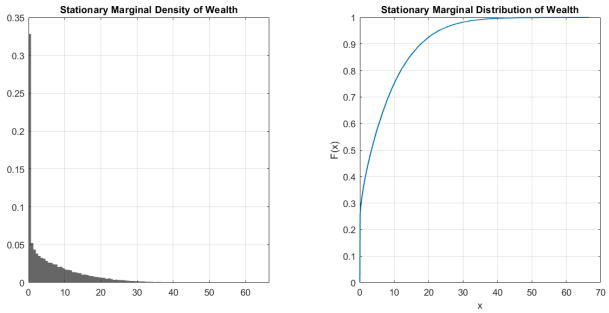

Figure 7: Stationary Marginal Distribution of Wealth, $\gamma = 2$, EIS $= 0.1$

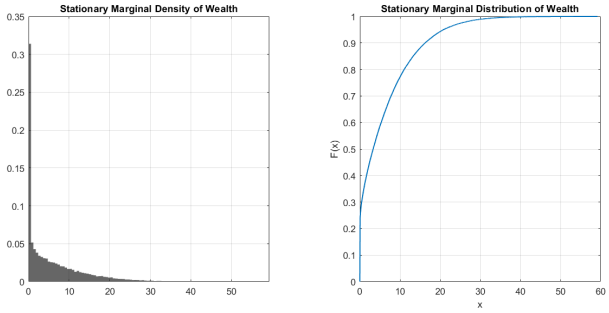

Figure 8: Stationary Marginal Distribution of Wealth, $\gamma = 2$, EIS $= 0.2$

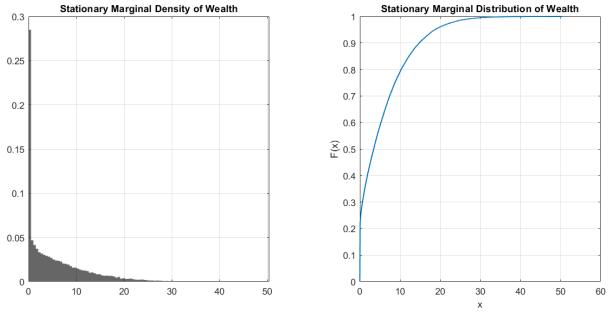

Figure 9: Stationary Marginal Distribution of Wealth, $\gamma = 2$, EIS $= 0.8$

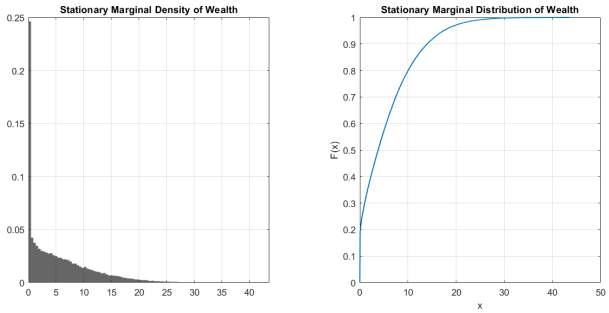

Figure 10: Stationary Marginal Distribution of Wealth, $\gamma = 2$, EIS $= 2$

Code Readme

All code is created and executed with MATLAB R2022b and may not run properly on older versions of MATLAB. Please also refer to published code (pdf) files to see the output. The code files are as follows:

File	Description
`aiyecon.m`	This function solves for the assets policy function of the incomplete markets model using CRRA preferences. It uses the EGM algorithm to iterate on the policy function using the Euler equation.
`aiyecon_epz.m`	This function solves for the assets policy function of the incomplete markets model using EPZ preferences. It uses the EGM algorithm to iterate on the value function using the first order condition.
`hw5.m`	This script simulates the markov chain, solves for the stationary general equilibrium and computes the stationary distribution and it's moments.
`rouwenhorst.m`	This function discretizes an AR(1) process using Rouwenhorst's method as in Kopecky and Suen (2010).
`der.m`	This custom written function computes the two-sided numerical derivative.